

KASHAN, Iran, Islamic Republic of

WMO: 407850

Lat: 33.9667N

Lon: 51.4801E

Elev: 982

StdP: 90.07

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.2	-2.1	-13.0	1.4	6.8	-10.9	1.7	7.3	8.3	11.4	6.2	10.4	0.2	210	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.7	42.2	19.1	41.0	18.9	39.8	18.6	21.6	38.3	20.6	37.5	19.8	36.7	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.5	12.4	29.8	14.0	11.2	29.5	12.9	10.5	28.4	67.7	38.3	64.1	37.6	61.0	36.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	6.5	5.3	-6.5	44.7	3.3	1.4	-8.9	45.7	-10.8	46.5	-12.7	47.3	-15.1	48.2		
			-7.1	23.2	2.7	2.3	-9.1	24.9	-10.7	26.2	-12.3	27.5	-14.2	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	5.2	8.2	13.8	19.0	25.1	30.8	33.4	31.9	27.2	20.6	12.0	6.9
	DBStd	10.29	3.92	3.75	4.10	3.62	3.11	2.53	2.17	1.95	2.60	3.44	3.74	2.88
	HDD10.0	356	151	68	12	1	0	0	0	0	0	0	21	103
	HDD18.3	1439	407	284	149	34	1	0	0	0	0	18	193	355
	CDD10.0	3846	3	18	131	271	468	623	724	678	516	328	79	6
	CDD18.3	1889	0	0	9	54	211	373	466	419	266	88	2	0
	CDH23.3	27611	0	0	124	558	2504	5730	7756	6666	3374	879	21	0
	CDH26.7	17034	0	0	33	164	1222	3666	5357	4409	1859	322	2	0
Wind	WSAvg	1.2	0.7	0.9	1.4	1.5	1.8	1.8	1.9	1.6	1.0	0.8	0.6	0.5
Precipitation	PrecAvg	141	22	16	25	22	13	2	0	1	0	4	16	17
	PrecMax	221	40	68	62	123	63	12	4	10	2	28	57	54
	PrecMin	73	1	0	0	0	0	0	0	0	0	0	0	2
	PrecStd	37	14	14	17	23	13	3	1	2	1	7	15	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	21.0	30.5	33.1	37.5	42.4	44.2	43.0	39.1	34.2	25.6	18.9
		MCWB	8.5	9.6	13.1	15.4	17.2	18.6	19.5	19.4	17.8	16.2	12.3	8.9
	2%	DB	15.2	18.5	26.2	30.2	35.8	40.8	42.9	41.4	37.5	32.0	23.0	16.7
		MCWB	7.3	8.5	11.7	14.5	16.7	18.3	19.1	19.1	17.4	15.1	11.7	8.2
	5%	DB	13.4	16.8	23.9	28.3	34.2	39.6	41.7	40.2	36.1	30.1	20.8	14.5
		MCWB	6.3	7.7	11.0	13.8	16.7	17.9	19.0	18.8	17.2	14.8	10.9	7.2
	10%	DB	11.7	15.1	21.7	26.6	32.7	38.2	40.3	39.0	34.8	28.1	18.8	12.5
		MCWB	5.5	7.0	10.1	13.2	16.1	17.6	19.0	18.5	16.9	14.3	10.0	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	10.8	13.9	17.0	20.4	22.6	23.0	22.6	20.2	17.7	14.0	10.0
		MCDB	15.6	17.1	26.6	28.2	33.3	37.8	39.6	38.6	35.5	30.4	22.3	15.9
	2%	WB	8.1	9.4	12.7	15.7	18.4	20.7	21.6	21.2	19.1	16.5	12.7	8.9
		MCDB	13.7	16.0	23.6	26.6	32.5	35.7	39.4	37.8	34.5	28.5	20.5	14.5
	5%	WB	7.0	8.5	11.7	14.8	17.5	19.4	20.7	20.2	18.2	15.6	11.7	8.1
		MCDB	12.1	15.2	21.1	25.3	31.2	35.2	38.5	37.3	33.6	27.2	18.6	12.5
	10%	WB	5.9	7.6	10.8	14.1	16.7	18.5	19.9	19.5	17.5	14.8	10.8	7.3
		MCDB	10.7	14.1	20.0	23.9	30.0	35.1	37.5	36.7	32.7	26.0	17.2	11.2
Mean Daily Temperature Range	5% DB	MDBR	10.3	11.3	12.2	12.4	13.1	14.0	13.7	14.7	14.6	13.3	11.0	10.2
		MCDBR	13.1	13.6	15.3	14.8	14.6	15.3	15.1	15.8	15.9	15.5	13.5	13.3
		MCWBR	7.1	6.7	6.2	5.3	4.9	5.0	5.0	5.6	5.9	6.5	6.2	7.0
	5% WB	MCDBR	11.9	12.1	13.3	12.6	13.8	13.9	13.9	14.4	14.6	13.9	11.6	10.9
		MCWBR	6.9	6.4	5.9	5.1	5.4	5.6	5.6	6.0	6.2	6.4	5.6	6.0
Clear-Sky Solar Irradiance	taub	0.329	0.371	0.443	0.541	0.545	0.535	0.570	0.508	0.447	0.440	0.377	0.328	
	taud	2.327	2.184	1.966	1.774	1.781	1.814	1.757	1.892	2.021	2.019	2.190	2.355	
	Ebn at Noon	872	865	825	762	764	769	739	780	811	776	800	849	
	Edh at Noon	100	128	174	221	221	214	225	193	162	150	114	92	
All-Sky Solar Radiation	RadAvg	3.04	3.99	5.16	5.94	7.01	8.13	7.79	7.31	6.24	4.69	3.34	2.79	
	RadStd	0.26	0.24	0.35	0.42	0.38	0.27	0.24	0.19	0.18	0.22	0.27	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.53	N/A	N/A	+0.98	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	-1.23	-1.34	N/A	N/A	N/A	N/A

Nomenclature: See separate page